



Cannabis pathogens X: *Phoma*, *Ascochyta* and *Didymella* species

J. M. McPartland

To cite this article: J. M. McPartland (1994) *Cannabis* pathogens X: *Phoma*, *Ascochyta* and *Didymella* species, *Mycologia*, 86:6, 870-878, DOI: [10.1080/00275514.1994.12026492](https://doi.org/10.1080/00275514.1994.12026492)

To link to this article: <https://doi.org/10.1080/00275514.1994.12026492>



Published online: 29 Aug 2018.



Submit your article to this journal [↗](#)



Article views: 2



View related articles [↗](#)



Citing articles: 3 View citing articles [↗](#)

Cannabis pathogens X: *Phoma*, *Ascochyta* and *Didymella* species

J. M. McPartland

Vermont Alternative Medicine/AMRITA, 53
Washington St., Middlebury, Vermont 05753

Abstract: Eight species of *Phoma* and *Ascochyta* are described and contrasted. *Diplodina cannabicola*, *Diplodina parietaria* f. *cannabina*, *Erysiphe communis* var. *urticirum*, *Phyllosticta cannabis*, and *Ascochyta cannabis* become synonyms of *Phoma cannabis*, comb. nov. ($\equiv$ *Depazea cannabis*). Other *Phoma* species on *Cannabis* include *P. exigua* ($\equiv$ *P. herbarum* f. *cannabis*, *Plenodomus cannabis*), *P. herbarum*, *P. glomerata*, and *P. piskorzii*. *Ascochyta arcuata*, sp. nov. is described as the anamorph of *Didymella arcuata*. *Ascochyta prasadii* and *Ascochyta cannabina* are also described. *Didymella arcuata* is contrasted with *D. cannabis*. The pathogen complex is compared to one involving a related host, stinging nettles, *Urtica dioica*. Host range and geographic distribution of the eight fungi are discussed.

Key Words: *Ascochyta*, *Cannabis*, *Didymella*, *Phoma*

INTRODUCTION

A number of fungi purportedly cause brown leaf spot and stem canker of *Cannabis* (McPartland, 1991). They include species of *Phyllosticta*, *Ascochyta*, *Phoma* and *Diplodina*. Mycologists named these species during the Saccardoan era, placing great emphasis on host, substrate, and conidial septation. This contrived system led to a chaotic situation where plurivorous fungi obtained synonyms in all four genera (Boerema and Bollen, 1975).

Schultes et al. (1974) described *Cannabis* as a “neglected taxon.” Its pathogens have fared a similar fate. Brown leaf spot and stem canker of *Cannabis* is ubiquitous, second only to yellow leaf spot (caused by *Septoria* spp.) among herbarium collections. The two diseases occur sympatrically, sometimes on the same plant. The literature cites 23 different taxa causing brown leaf spot and stem canker of *Cannabis*. These 23 taxa are reassessed and transferred where needed.

MATERIALS AND METHODS

Methods for downloading taxonomic literature and locating type materials are previously described (McPartland, 1994a). Common names of diseases follow those adopted by the American Phytopathological Society (McPartland, 1991). Studying *Phoma* spp. by their host range parallels Farr’s (1991) approach to the study of *Septoria* spp. These compilations and comparisons are especially useful when cultures are lacking.

Conidiomata were dislodged from leaf or stem tissue with dissecting needles, rehydrated in 3% KOH, then measured at their widest part. They were either hand-sectioned with a razor blade or teased apart with needles under a 4 $\times$ objective. Material was stained with either cotton blue in lactic acid or 1% phloxine in water (phloxine subsequently replaced with KOH by capillary action under a cover slip). Curved conidia were measured along their circumference, widths measured at their widest part. Measurements are listed as a range; for ample material ($n \geq 50$), mean measurements are given with extremes in brackets.

Cultures techniques follow those of Boerema and Bollen (1975). Fungi were plated on Difco oatmeal agar and/or potato dextrose agar at room temperature (20–22 C), exposed to mixed incandescent and fluorescent (UV-B) light, approx. 70 W/m² for 12 h/day for 7 days, at ambient humidity (approx. 30% RH). Pigments from fungal metabolites diffusing into agar were treated with dilute NaOH, approx. 1 N solution.

DESCRIPTION OF THE SPECIES

The 23 different taxa appearing in the literature are distributed among eight species. Their morphology, taxonomy and nomenclature are described.

Didymella cannabis (Winter) von Arx in Müller and von Arx, Beitr. Kryptogamenfl. Schweiz 11(2): 365. 1962.

FIG. 1

$\equiv$ *Sphaerella cannabis* Winter, Hedwigia 11(10): 145. 1872.

$\equiv$ *Mycosphaerella cannabis* (Winter) Magnus, in Dalla Torre & Sarnthein, Fl. Tirol, Vorarlberg, Liechtenstein, III, Pilze, p. 465. 1905.

$\equiv$ *Mycosphaerella cannabis* (Winter) Röder, Z. Pflanzenkrankh. 47(10): 530. 1937.

ANAMORPH. *Phoma cannabis* (L. Kirchner) McPartland, *comb. nov.* FIGS. 2–5

- = *Depazea cannabis* L. Kirchner, Lotos 6: 183, 1856.
- = *Phyllosticta cannabis* (L. Kirchner) Spegazzini, Atti Soc. Crittog. Ital. 3: 67, 1881, in error as "(Lasch) Speg."
- = *Ascochyta cannabis* (Spegazzini) Voglino, Ann. R. Acad. Agric. Torino 55: 199, 1913, non *Ascochyta cannabis* Lasch, Klotzchii Herb. vivum exsic., Ed. I, no. 1509, 1846 (= *Septoria cannabis* (Lasch) Sacc.)
- = *Erysiphe communis* var. *urticirum* Westendorp, Bull. Acad. Roy. Sci. Belgique 21: 231, 1854.
- = *Diplodina cannabicola* Petrak, Ann. Mycol. 19: 122, 1921.
- = *Diplodina parietaria* Brun. f. *cannabinata* von Höhnelt, Verh. Zool.-Bot. Ges. Wien 60: 314, 1910.

Colonies (in vitro) not described. Pseudothecia dark brown, immersed then erumpent, flattened on host substrate (stems), globose in culture, (90–)135(–180) μm diam on host substrate (to 190 μm in culture), averaging 140 asci per pseudothecium (Röder, 1937). Ostioles papilliform. Asci emerging parallel from hymenium, bitunicate, clavate to cylindrical, with a slight pedicle, eight-spored, 50–85 $\times$ 9–10 μm . Ascospores hyaline, subovoid to oblong, submedially one-septate with two cells unequal in size, constricted at septum, (11–)13.6(–17) $\times$ (4.5–)5.4(–8) μm . Pseudoparaphyses filamentous, 0- to 3-septate, 25–50 $\times$ 1 μm .

Pycnidia brown-black, immersed then erumpent, globose to subglobose, peridium three to four cells thick (5–10 μm), *textura angularis-globulosa*, (65–)130(–180) μm diam on host substrate. Ostioles round, sometimes slightly papillate. Conidiogenous cells enteroblastic, phialidic, ampulliform to short cylindrical, 2–6 μm tall. Conidia hyaline to light brown en masse, oval to ellipsoidal, sometimes biguttulate, 0- to 1-septate on host (nonseptate in culture), 3–8 $\times$ 2–3 μm on host substrate (3–10 $\times$ 2–4 μm in culture), adhering to one another to form densely aggregated masses. Chlamydospores intercalary or terminal, usually globose to oval (in culture sometimes a dictyochlamydospore), single or in chains, brown, with smooth thick walls, 8–17 μm diam.

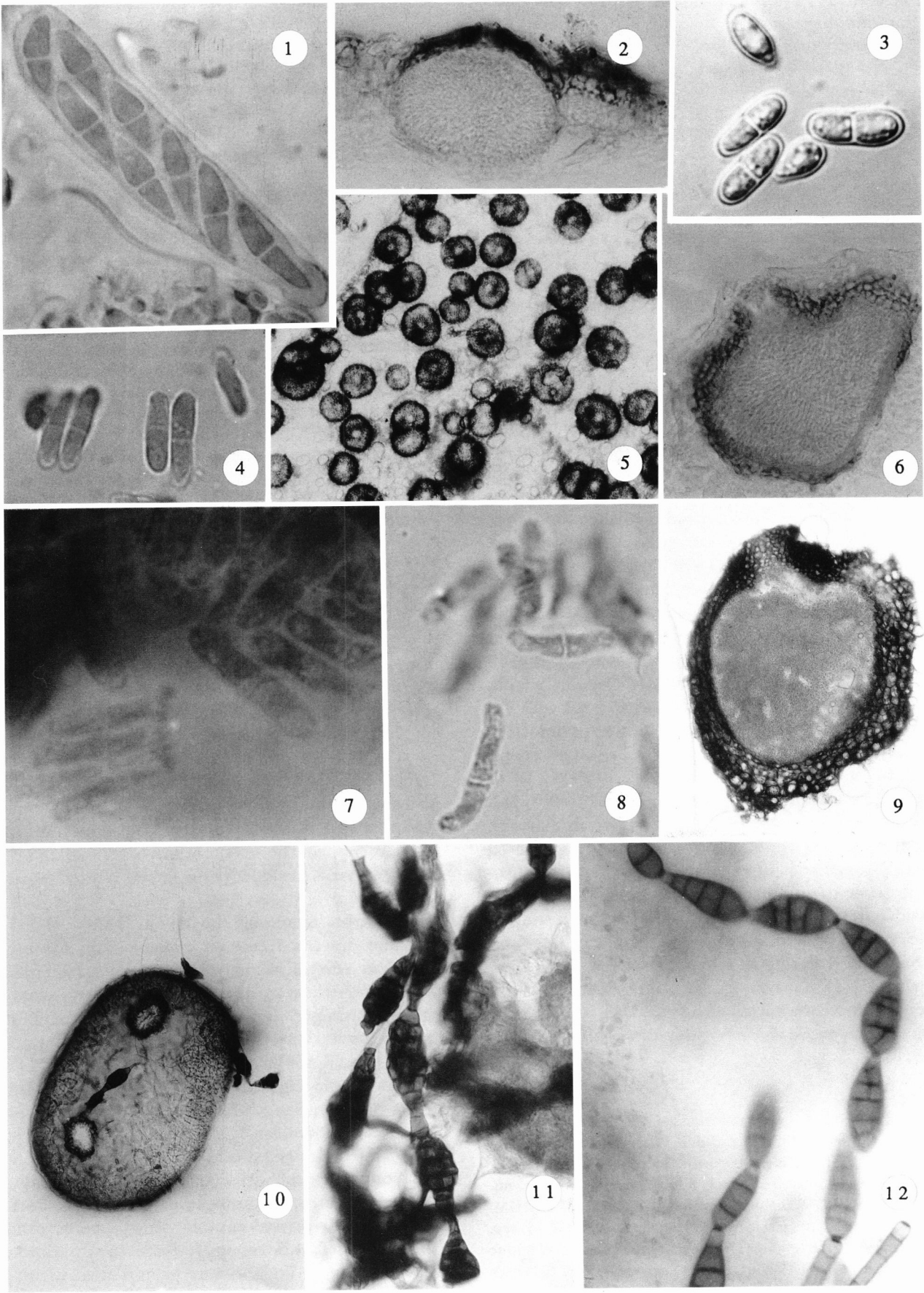
NEOTYPE designated herein: GERMANY. Giesen, on *C. sativa* leaves, leg. Hofmann, det. Rabenhorst, Herbarium vivum mycologicum, Ed. II, no. 559, labeled *Spilosphaeria cannabis* (= *Depazea cannabis*)—1857 (BPI).

ADDITIONAL SPECIMENS EXAMINED. GERMANY. Berlin, on *C. sativa* stems, 1936, leg. and det. Röder, labeled *Didymella cannabis* (B). UKRAINE. Stryj (now Stry), on *C. sativa* stems, 1-IX-1917, leg. and det. Petrak, Fungi polonici exsiccati no. 580, SYNTYPE of *Diplodina cannabicola* (BPI). BELGIUM. Louvain, on *C. sativa* leaves and female flowers, 1854, leg. Maertens, det. Westendorp, Herbar Cryptogamique Belgique no. 972, LECTOTYPE of *Erysiphe communis* var. *urticirum*

designated herein (BR). ROMANIA. Ciornuleasa (Ilfov), on "*C. ruderalis*" (actually *C. sativa*) leaves, 18-VI-1971, leg. Negrean, det. Constantinescu, Herbarium Mycologicum Romanicum no. 2473, labeled *Ascochyta cannabis* (BPI). LATVIA. Riga, on *C. sativa* stems, 1932, leg. and det. Starcs, labeled *Didymella cannabis* (B). ITALY. Montello (Treviso), on *C. sativa* leaves, IX-1901, leg. and det. Saccardo, Mycotheca italica no. 936, labeled *Phyllosticta cannabis* (BPI). ITALY. On *C. sativa* leaves, IX-1901, leg. and det. Ciferri, labeled *Phyllosticta cannabis* (BPI). POLAND. On *C. sativa* leaves, undated, leg. and det. Raciborski, Mycotheca polonica no. 188, labeled *Phyllosticta cannabis* (BPI). SERBIA. Drina, on *C. sativa* leaves, 21-VIII-1909, leg. Ranojevic, det. Kabát and Bubák, Fungi imperfecti exsiccati no. 601, labeled *Phyllosticta cannabis* (BPI). NEPAL. Kathmandu, on *C. indica* stems, XI-1986, leg. & det. McPartland, Mycothecum loannes no. 86-23, labeled *Didymella cannabis* (BPI); Hille, on *C. indica* leaves, XI-1986, leg. and det. McPartland, Mycothecum loannes no. 86-19, labeled *Phoma cannabis* (BPI); Jomson, on *C. indica* stems, XI-1986, leg. and det. McPartland, Mycothecum loannes no. 86-25, labeled *Phoma cannabis* (BPI). INDIA. 3 km N of Chandigarh, on *C. indica* leaves, 28-VI-1992, leg. and det. McPartland, Mycothecum loannes no. 92-3, labeled *Phoma cannabis* (BPI). PAKISTAN. Kagan Valley, on *C. indica* leaves, 13-VIII-1975 leg. & det. Basit, labeled *Phyllosticta cannabis* (BPI). USA. ILLINOIS: Oquawka, on *C. sativa* leaves, VIII-1924, leg. and det. Young, labeled *Phyllosticta cannabis* (ILLS). MARYLAND: Yarrow, on *C. sativa* leaves, 14-VIII-1916, leg. J. T. Rogers, det. G. R. Lyman, labeled *Phyllosticta cannabis* (BPI); Yarrow, on *C. sativa* leaves, 8-IX-1916, leg. G. R. Lyman, det. J. T. Rogers, labeled *Phyllosticta cannabis* (BPI). KANSAS: Troy, on *C. sativa* leaves, 4-IX-1952, leg. & det. W. H. Horr, labeled *Phyllosticta cannabis* (BPI). WISCONSIN: Brodhead, on *C. sativa* leaves, 4-IX-1957, leg. Greene, det. Cash, labeled *Phyllosticta cannabis* (BPI).

Note. *Sphaerella cannabis* Winter was originally described from hemp stems in Germany. No anamorph was described. Röder (1937) used single ascospore cultures to connect the teleomorph with the anamorph. Müller and von Arx (1962) transferred Winter's fungus to *Didymella*.

Anamorph taxonomy begins in 1856, when L. Kirchner described *Depazea cannabis*. This name has served in several recombinations. Spegazzini (1881) proposed *Phyllosticta cannabis* (Lasch) Speg. without citing a basionym. It is evident from the description that Spegazzini (1881) gave for that fungus under number 150 could not have been *Ascochyta cannabis* Lasch because of the big differences in conidial measurements between Spegazzini's collection and the protologue of *A. cannabis*. It is likely that Spegazzini had intended *Depazea cannabis* L. Kirchner, the description of which agrees well with Spegazzini's fungus, and was thus in error in citing Lasch. Saccardo (1884) corrected Spegazzini's error, and was followed by several authors (Voglino, 1913; Röder, 1937; Farr, 1973). *Ascochyta cannabis* (Speg.) Voglino was proposed in error by



Vogolino (1913) in reference to *Depazea cannabis* L. Kirchner.

The holotype of *Depazea cannabis* is missing from B (pers. comm., B. Hein, 1991). A specimen at BPI is here designated as a neotype; it is labeled *Spilosphaeria cannabis* Rabenhorst 1857 (Herbarium vivum mycologicum, Ed. II, no. 559). The printed label lists *D. cannabis* in synonymy under *Sp. cannabis*. The specimen contains brown leaf spots with pycnidia that match Kirchner's description. Conidia and conidiogenous cells therein suggest a *Phoma* species. I transfer Kirchner's taxon to that genus, as *P. cannabis*.

Unfortunately, the neotype also exhibits yellow leaf spots, containing *Septoria cannabis* (Lasch) Saccardo. Note that specimen no. 559 at B contains only *S. cannabis* and no *P. cannabis*. Also, the label at B is different; there is no mention of *D. cannabis* in synonymy. Stevenson (1967) has previously documented substitutions in Rabenhorst's series of exsiccati. While new taxa distributed as specimens in exsiccatae are isotypes (each one a potential lectotype), Rabenhorst's substitutions are not isotypes. Thus, the part at BPI is a neotype of *P. cannabis*, not a lectotype. The part at B is designated a lectotype of *Spilosphaeria cannabis*, a synonym of *S. cannabis* (McPartland, 1994c).

Two taxa of *Diplodina* Westendorp are also synonymized, *Diplodina parietaria* Brun. f. *cannabina* von Höhnelt and *Diplodina cannabicola* Petrak. They were described from Germany and the Ukraine, respectively. Petrak (1923) examined von Höhnelt's fungus and synonymized it under *D. cannabicola*. A syntype of *D. cannabicola* at BPI contains conidia averaging 5 µm long, often guttulate, with 0–1 septa. Its simple conidiophores and ostiolate pycnidia preclude it from Sutton's (1980) concept of *Diplodina*.

Petrak (1923) suggested *D. cannabicola* may be identical to *Dendrophoma marconii* Cavara. Brandenburger (1985), following Petrak, synonymized the two species. Petrak probably examined a specimen of *D. marconii* distributed by Briosi and Cavara (I Funghi Parassiti delle Piante Coltivate od Utili no. 20). Specimen no. 20 at BPI is an *Ascochyta* species, quite different from Cavara's type of *D. marconii* (PAV!). Briosi and Cavara, like Rabenhorst, must have substituted fungi in their exsiccata.

Oudemans (1920) cited "*Erysimum commune* Duby, i

Urticearum West., Not. IV, p.6." This multiple orthographic error (three spelling errors, wrong author, and mistaken page number) represents *Erysiphe communis* var. *urticirum* Westendorp. Papers accompanying Westendorp's type indicate that he first identified the fungus as "*E. martii* Lev." (Léveillé, *nomen rejiciendum*), then changed to *E. c.* var. *urticirum*. The specimen at BR is covered by a fine white meshwork, consisting of elongate leaf trichomes and spider mite webbing. There is no powdery mildew. Fruiting bodies in brown leaf spots are ostiolated, a *Phoma* species, and identical to Kirchner's taxon. Although Westendorp's varietal epithet antedates the species *P. cannabis*, it cannot have priority outside of its rank (see Art. 60, Greuter et al., 1988).

Phoma cannabis often produces septate conidia in vivo (a "pseudo-*Ascochyta*," *vide* Boerema and Bollen, 1975). In vivo formation of septate conidia occurs in many *Phoma* species (e.g., *P. exigua* Desm., = *A. phaseolorum* Sacc.), where septation is not seen in vitro (Boerema and Bollen, 1975) Röder (1939) and von Arx (in Müller and von Arx, 1962) cultured *P. cannabis* and recovered only nonseptate conidia. The only known host of *P. cannabis* is *Cannabis*. Herbarium specimens are limited to the northern hemisphere.

Didymella arcuata Röder, Phytopathol. Z. 12: 329. 1939.

Pseudothecia brown-black, immersed or erumpent, globose on leaves and flattened on stems, 130–150 µm diam on host (to 250 µm in vitro). Ostioles round. Asci emerging parallel from hymenium, bitunicate, clavate to pyriform, with a slight pedicle, eight-spored, 50–80 × 9–14 µm. Ascospores hyaline, subovoid to oblong, submedially one-septate with two cells unequal in size, constricted at septum, (16–)18.6(–22) × (4–)5(–6) µm. Pseudoparaphyses filamentous, 0- to 3-septate, 23–60 × 0.5–1.5 µm.

ANAMORPH. *Ascochyta arcuata* McPartland, *sp. nov.* FIGS. 6–8

Coloniae (in vitro, ex Abkulturae) ad marginem laeves, ex albo glaucae cum lilacinae, reversae cum pigmento griseo-rubro effluenti circumdatae; mycelio aërio abundanti, pycnidiis sparsis. Pycnidia (in vivo) atrobrunnea, immersa tum

FIGS. 1–12. Fungi on *Cannabis*. 1–5. *Didymella cannabis-Phoma cannabis*. 1. Ascus and ascospores, ×1240. 2. Pycnidium, ×300. 3. Conidia, ×1680. 4. Conidia from syntype of *Diplodina cannabicola*, ×1720. 5. Pycnidia on leaf surface, from lectotype of *Erysiphe communis* var. *urticirum*, ×100. 6–8. *Ascochyta arcuata*. 6. Pycnidium, ×300. 7. Conidia in cotton blue, ×1650. 8. Conidia in phloxine, ×1650. 9. *Phoma exigua* pycnidium from syntype of *Plenodomus cannabis*, ×160. 10, 11. *Phoma glomerata*. 10. Double-ostiolated pycnidium, ×150. 11. Chain of chlamydospores, ×490. 12. *Alternaria alternata* conidiophores and chains of conidia, ×660.

erumpens, subglobosa, unilocularia, ostiolata, textura angulari-globulosa, (90–)135(–150) μm lata in vivo (usque ad 450 μm in vitro). Cellulae conidiogenae enteroblasticae, phialidicae, discretae, determinatae, simplicae, ampulliformes vel brevicylindricae, (5–)8(–11) μm longitudini. Conidia hyalina, dilute brunnea aggregata, ellipsoidea vel brevicylindrica, recta vel curvata, plerumque 1-septata, raro 2-septata, ad septum non constricta, subtiliter guttulate, 9–28 $\times$ 3.5–8.0 μm in vivo (accrescentia in vitro). Chlamydosporae raras, intercalariae, globosae vel ovaes, singulares vel brevicatenatae, atrobrunneae, glabrae, 8–12 μm latae.

Colonies (in vitro, from Abkultures) with even margins, white turning gray-green with purple hues, reverse with reddish pigment diffusing into agar, turning sepia-brown with the application of NaOH; producing much aerial mycelium but few pycnidia. Pycnidia (in vivo) dark brown, immersed then erumpent, subglobose, unilocular, ostiolated, peridium *textura angularis-globulosa*, (90–)135(–150) μm diam on host substrate (up to 450 μm diam in culture). Conidiogenous cells enteroblastic, phialidic, discrete, determinate, ampulliform to short cylindrical, (5–)8(–11) μm tall. Conidia hyaline, light brown en masse, ellipsoidal to short cylindrical, straight or curved, usually one-septate (rarely two-septate), not constricted at the septum, finely guttulate, 9–28 $\times$ 3.5–8.0 μm on host substrate; larger in culture (to 35 $\times$ 9 μm in vitro) *vide* Röder (1939). Chlamydospores rare, intercalary, globose to oval, single or in short chains, brown, smooth, 8–12 μm diam.

ANAMORPH HOLOTYPE designated herein: ITALY. Forlì et Pavia, on *C. sativa* stems, IX-1887, and VIII-1888, leg. and det. *Briosi and Cavara*, I Funghi Parassiti delle Piante Coltivate od Utili no. 20, labeled *Dendrophoma marconii* (BPI 0381272).

ADDITIONAL COLLECTIONS EXAMINED. GERMANY. Berlin, isolated from *C. sativa* leaves, 1940, leg. and det. Röder, living culture at CBS 118.40; 1991 subculture at ATCC 90203 (dried *abkulturs* deposited at BPI).

OTHER SPECIES EXAMINED. ITALY. Forlì, on *C. sativa* stems, IX-1887, leg. and det. *Cavara*, HOLOTYPE of *Dendrophoma marconii* (PAV). GERMANY. Driesen, on *C. sativa* leaves, 1846, leg. and det. *Lasch*, Rabenhorst's Klotzschii Herbarium vivum mycologicum, Edition Nova no. 1509, labeled *Ascochyta cannabis*, LECTOTYPE of *Septoria cannabis* (B); Driesen, on *C. sativa* leaves, 1846, leg. and det. *Lasch*, Rabenhorst's Klotzschii Herbarium vivum mycologicum, Edition Nova no. 1509, labeled *Ascochyta cannabis* (BPI).

Note. Röder (1939) described *D. arcuata* as a new species, producing an *Ascochyta* anamorph in vivo and in vitro. He considered the anamorph to be *Ascochyta cannabis* Lasch, even though Saccardo (1884) transferred the taxon to *Septoria* as *S. cannabis* (Lasch) Sacc. Röder (1939) reported two fungi on Lasch's specimen. One was a *Septoria*; the other he interpreted as an

Ascochyta. Röder retained the species in *Ascochyta*, rejecting Saccardo's recombination. I disagree on three points: first, the protologue includes a description of elongate, *Septoria*-like conidia; second, *S. cannabis* is firmly entrenched in common usage, whereas *A. cannabis* is not; third, only one tiny leaf spot on the holotype specimen contains a non-*Septoria* fungus, and it is not an *Ascochyta*, but a *Phoma* species. Rejecting *A. cannabis* Lasch requires a new taxon for the anamorph of *D. arcuata*. Brandenburger (1985) noted this difficulty, but simply labeled the *D. arcuata* anamorph as "*Ascochyta* sp."

Röder deposited his types at herb B (pers. comm., K. Röder, 1986); the type of *D. arcuata* is missing (pers. comm., B. Hein, 1987). Röder (1939) designated an ex type culture, CBS 118.40. Unfortunately, it has degraded to an *Abkultur*, and no longer produces pycnidia. I sent a subculture to ATCC; workers there were equally unsuccessful (pers. comm., S. C. Jong, ATCC, 1993). Lastly, sterilized hemp stems on water agar were inoculated with CBS 118.40, but again pycnidia did not form. These cultures remain at ATCC and CBS for future DNA studies.

A specimen at BPI may serve as a neotype for the anamorph: Funghi Parassiti delle Piante Coltivate od Utili no. 20. This specimen, by Briosi and Cavara, is labeled *Dendrophoma marconii* Cavara. Its morphology agrees with that of the anamorph described by Röder (1939). The BPI specimen is not congeneric with Cavara's type of *D. marconii*, which is something else (McPartland, 1994b).

According to Morgan-Jones et al. (1991), conidium length is the best criterion for distinguishing *Ascochyta* from *Phoma*. In vivo measurements herein demonstrate that *A. arcuata* conidia are longer than those of *P. cannabis*. This has also been documented in vivo by Röder (1937, 1939). Boerema and Bollen (1975) separated *Phoma* and *Ascochyta* on the basis of in vitro septation and conidiogenesis. Buchanan (1987) did not see the differences in conidiogenesis. Conidia of *A. arcuata* are septate in vitro; conidia of *P. cannabis* are unicellular (Röder, 1937, 1939). There is no difference in conidiogenesis between *A. arcuata* and *P. cannabis*. The only known host of *A. arcuata* is *Cannabis*, and it only occurs in Europe.

Phoma exigua Desmazières, Ann. Sci. Nat. Bot., Ser. 3, 11: 282. 1849. FIG. 9

= *Ascochyta phaseolorum* Saccardo, Michelia 1(2): 164. 1878.

= *Plenodomus cannabis* (Allescher) Moesz & Smarods in Moesz, Bot. Közl. 38: 70. 1941.

≡ *P. herbarum* f. *cannabis* Allescher, Rabenhorst's KryptogFlora, Pilze 6: 330. 1899.

TELEOMORPH. None known.

Colonies (in vitro) with a lobed or scalloped border, grayish-white, producing dense, felty aerial mycelium but sparse pycnidia. Underside unpigmented, turning blue-green then brown-red with the application of NaOH. Pycnidia immersed but eventually erumpent, solitary or gregarious, shiny brown-black, globose, pseudoparenchymatous, peridium five to seven cells thick, *textura angularis-globulosa*, 110–220 μm diam. Ostioles slightly papillate, 10–60 μm diam. Conidiogenous cells enteroblastic, phialidic, ampulliform, hyaline. Conidia hyaline, ellipsoid or cylindrical, straight or slightly curved, often biguttulate and becoming one-septate, $5\text{--}10 \times 2\text{--}3.5 \mu\text{m}$. Chlamydospores not present.

COLLECTIONS EXAMINED. LATVIA. Riga, on *C. sativa* stems, V-1939, leg. Smarods, det. Moesz and Smarods, Petrak's Mycotheca generalis no. 1870, SYNTYPE of *Plenodomus cannabis* (BPI); Riga, on *C. sativa* stems, V-1939, leg. Smarods, det. Moesz and Smarods, Petrak's Mycotheca generalis no. 1870, labeled *Plenodomus cannabis* but not marked "type" (BPI).

Note. Boerema (1970) examined Allescher's type of *P. herbarum* f. *cannabis* and reduced it to synonymy of *P. exigua* (*Phoma* section *Phyllostictoides*). My study of Moesz and Smarods' material reveals thick pycnidial walls that do not stain with Lugol's iodine, indicative of *Phoma* section *Plenodomus*, Category A, *fide* Boerema et al. (1981).

Boerema (1972) considered *A. phaseolorum* to be a synonym of *P. exigua*. Sutton and Waterston (1966) listed *Cannabis* as a host of *A. phaseolorum*. But measurements they gave for conidia ($13\text{--}16.5 \times 3.5\text{--}5 \mu\text{m}$) are too large for *P. exigua*. Domsch et al. (1980) considered *P. exigua* the most common pycnidial fungus in the world. It is cosmopolitan, attacking over 200 plant genera, and has more than 100 synonyms. For a more detailed description of *P. exigua*, see Morgan-Jones and Burch (1988).

Phoma glomerata (Corda) Wollenweber & Hochapfel, Z. Parasitenk. (Berlin) 8: 592. 1936. FIGS. 10, 11

TELEOMORPH. None known.

Colonies olivaceous-gray with sparse aerial mycelium except sectors producing fluffy growth bearing pycnidia and chlamydospores. Underside of colony pale to dark olivaceous-gray. Pycnidia dark brown, erumpent, globose to obpyriform, peridium three to five cells thick (10–12 μm) and *textura angularis-globulosa*, 50–300 μm diam. Ostioles round, often papillate. Conidiogenous cells enteroblastic, phialidic, ampulliform to doliiform. Conidia hyaline to pale olive-brown, ellipsoid to short cylindrical, straight or slightly curved, guttulate, unicellular or rarely one-septate, $4.5\text{--}10 \times$

$1.5\text{--}4 \mu\text{m}$. Chlamydospores terminal, *Alternaria alternata*-like in shape with many transverse and longitudinal septa, single or in chains, brown, thick walled, smooth, become roughened with age, $30\text{--}65 \times 15\text{--}20 \mu\text{m}$. Micropycnidia arising from mycelium or chlamydospores, often adjacent to pycnidia, globose, 20–50 μm diam.

COLLECTIONS EXAMINED. USA. ILLINOIS: Chicago, east of railroad tracks near Lake Shore Drive and South 47th St., on leaves of *C. sativa*-*C. afghani* hybrid, VII-1986, leg. and det. McPartland, Mycothecum loannes no. 86-7 (BPI).

Note. This collection represents a new host record. The plant was previously stressed by an unseen leaf-eating arthropod (probably *Psylliodes attenuata* (Koch)). It was also infected with *Alternaria alternata* (Fries: Fries) Keissler, conidia of which were confused with *P. glomerata* chlamydospores (cf. FIGS. 11, 12). Sutton (1980) reported *P. glomerata* from 94 plant genera and described it from air, sea, and humans, world-wide. He presented a more detailed description than the one herein.

Phoma herbarum Westendorp, Bull. Acad. Roy. Belgique Cl. Sci. 19: 118. 1852.

TELEOMORPH. None known.

Colonies (in vitro) with even margins, grayish-green in color, producing sparse aerial mycelium but abundant pycnidia. Colony reverse with diffusing sepia pigment, turning blue-green in NaOH. Pycnidia superficial, solitary or densely gregarious, simple or compound with two to three ostioles, usually globose (sometimes lenticular or flask-shaped), 80–260 μm diam (to 350 μm in vitro), walls pseudoparenchymatous, *textura angularis-globulosa*. Ostioles circular, non-papillate or slightly papillate, 15–30 μm diam. Conidiogenous cells enteroblastic, phialidic, ampulliform, hyaline, continuous with inner wall of pycnidia. Conidia hyaline (pinkish en masse), nonseptate, nonguttulate, ellipsoid, mostly $4.5\text{--}5.5 \times 1.5\text{--}2.5 \mu\text{m}$. Chlamydospores not present.

Note. This fungus reportedly causes stem cankers on *Cannabis* in Italy (Sydow, 1898), France (Brunaud, 1899), Denmark (Lind, 1913), the Netherlands (Oudemans, 1920), and Japan (Kyokai, 1965). Saccardo (1884) described conidia as $6\text{--}11 \times 3\text{--}4 \mu\text{m}$, which is too large for this species. No voucher specimens of *P. herbarum* on *Cannabis* could be located at B, BPI, BR, or MICH. The description above is extracted from the more detailed descriptions of Morgan-Jones (1988) and Boerema (1976). Boerema (1970, 1976) considered *P. herbarum* a weak pathogen, that Domsch et al. (1980) report from over 35 host genera, worldwide.

TABLE I. Comparison of eight *Phoma* and *Ascochyta* species reported from *Cannabis*¹

Species	Pycnidia (μm)	Conidia	Chlamydospores	Teleomorph
<i>P. cannabis</i>	(65-)130(-180)	Oval-ellipsoid, 3-8 × 2-3 μm, one to two celled, guttulate	Globose to alternarioid, 8-17 μm	<i>Didymella</i>
<i>A. arcuata</i>	(90-)135(-150)	Short-cylindrical, 9-28 × 3.5-8.0 μm, usually two celled, guttulate	Globose to oval, 8-12 μm	<i>Didymella</i>
<i>P. exigua</i>	110-220	Ellipsoid-cylindrical, 5-10 × 2-3.5 μm, one to two celled, guttulate	Absent	None known
<i>P. glomerata</i>	50-300	Ellipsoid-cylindrical, 4.5-10 × 1.5-4 μm, usually one celled, guttulate	Alternarioid, 30-65 × 15-20 μm	None known
<i>P. herbarum</i>	80-260	Ellipsoid, 4.5-5.5 × 1.5-2.5 μm, usually one celled, nonguttulate	Absent	None known
<i>P. piskorzii</i>	300-500	Ellipsoid, 6-16 × 2-3 μm, usually one celled, nonguttulate	Absent	<i>Leptosphaeria</i>
<i>A. prasadii</i>	55-93	Oblong, 7.5-10 × 1.5-2.5 μm, usually two celled, nonguttulate	Absent	None known
<i>A. cannabina</i>	120-130	Oval-oblong, 7-8.8 × 3.2-4 μm, usually two celled, nonguttulate	Absent	None known

¹ Characters taken from in vivo material.

Leptosphaeria acuta (Hoffman : Fries) P. Karsten, Mycologia Fennica 2: 98. 1873.

≡ *Sphaeria acuta* Hoffman, Vegetabilia cryptogama, Fascicle 1, p. 22. 1787.

≡ *Cryptosphaeria acuta* (Hoffman : Fries) Greville, Flora edinensis, p. 360. 1824.

?= *Leptosphaeria acuta* f. *cannabis* Roumeguère, Fungi selecti exsiccati no. 4172. 1887.

ANAMORPH. *Phoma piskorzii* (Petrak) Boerema & Loerakker, Persoonia 11: 315, 1981.

≡ *Diploplenodorus piskorzii* Petrak, Ann. Mycol. 21: 123. 1923, non *Phoma acuta* auct., nomen ambiguum.

Ascomorphs (in vivo) immersed becoming superficial, conic, surface smooth, black, wall scleroplectenchymatous, *textura prismatica*, 350-500 μm diam. Ostioles rounded, without periphyses, 30-40 μm diam. Asci emerging from broad hymenium, bitunicate, cylindrical, short stalked, eight-spored, 120-150 × 7-12 μm. Ascospores pale yellow, narrowly ellipsoidal, usually seven-septate, constricted at central septum, guttulate, smooth, 37-45 × 5-7 μm.

Pycnidia variable in shape but usually flattened-subglobose, papillate, to 500 μm diam, wall scleroplectenchymatous. Ostioles round, sometimes papillate. Conidia hyaline, nonguttulate, elliptical, unicellular but occasionally one-septate, (6-9(-16) × 2-3 μm. Chlamydospores absent.

COLLECTIONS EXAMINED. BELGIUM. No location given, on *C. sativa* stems, undated but ca 1879, leg. *Libert*, Reliquiæ Libertianæ no. 56 (BR—3 packets); Liège, on *C. sativa* stems, 1887, leg. *Libert*, Roumeguère's Fungi selecti exsiccati no. 4172 (BPI, FH).

Note. Description of the teleomorph is derived from Shoemaker (1984): the anamorph description is from Boerema and Loerakker (1981). Shoemaker (1984), Crane and Shearer (1991) and other authorities cited *Leptosphaeria acuta* as a specialized pathogen of stinging nettles, *Urtica dioica* L.

Saccardo and Roumeguère (1883) reported *Leptosphaeria acuta* on *Cannabis*. Roumeguère subsequently distributed *L. acuta* f. *cannabis* in his exsiccata. On the label, Roumeguère indicated the material came from another exsiccata, "Reliquiæ Libertianæ No. 191" (sic). Specimen no. 4172 from Roumeguère's exsiccata deposited at BPI is nearly effete, containing only one immature ascoma. Syntypes of *L. acuta* f. *cannabis* were not available at IMI, NYS, nor BR. BR, however, forwarded three specimens labeled *L. acuta*, from Reliquiæ Libertianæ no. 56. These only contain an anamorph, a *Jahniella*-like species with scleroplectenchymatous pycnidia. *Libert*'s original material needs to be located and inspected to confirm the synonymy with

L. acuta. Unfortunately, it is not indexed in her own exsiccata, Plantae Cryptogamicae quas in Arvenna (pers. comm., G. Cacavio, FH, 1994).

Many researchers (e.g., Shoemaker, 1984; Boerema and Loerakker, 1981) erroneously cited the authorities as *L. acuta* (Fuckel) P. Karsten. Crane and Shearer (1991) recognized Hoffman as the authority for the basionym.

Ascochyta prasadii Shukla & Pathak, Sydowia 21: 277. 1967.

TELEOMORPH. None known.

Colonies (in vitro) not described. Pycnidia globose to subglobose, light brown, thick walled, 55.8–93 × 52.7–86.8 µm. Ostioles round, 14.4–16 µm diam. Conidiogenous cells not described. Conidia hyaline, oblong, obtuse at the ends, one-septate, 7.5–10 × 1.5–2.5 µm. Chlamydospores absent.

Note. Shukla and Pathak isolated this fungus from *C. sativa* leaves in the botanical garden of the University of Udaipur, India. They named it a new species, “. . . characteristic of the genus *Ascochyta* not recorded on this host before,” overlooking *A. cannabis* Lasch and *A. cannabis* (Speg.) Voglino. It is difficult to assess the taxonomic position of *A. prasadii* because the holotype (IMI 118269) was not available.

Ascochyta cannabina E. I. Reichardt, Bull. Station Régionale Protectrice des Plantes à Leningrad 5: 46. 1925.

TELEOMORPH. None known.

Colonies (in vitro) not described. Pycnidia globose, brownish-yellow, sparse and irregularly arranged in leaf spots, 120–130 µm diam. Conidia hyaline, oblong-oval to subcylindrical, medially one-septate, hardly constricted at the septum, straight or slightly curved, 7–8.8 × 3.2–4 µm. Chlamydospores absent.

Note. Reichardt described leaf spots as irregular, whitish, with a pale gray-brown margin, often torn, occurring at the margins of leaves, 3–15 mm diam. He collected this fungus near Leningrad, on Aug. 28, 1921. The type is not deposited at LE (pers. comm., V. Mel'nik, 1993).

ADDITIONAL NOTES

In addition to the species described above, some researchers have simply cited their collections as “*Phoma* sp.” Srivastara and Naithani (1979) describe “*Phoma* sp.” on Indian *Cannabis* (conidia 2.4–3.1 × 1.4–1.6 µm). Paulsen (1971) isolated a *Phoma* sp. on wild hemp in Kansas. Brundza (1933), and Fuller and Norman

(1944, 1945) reported *Phoma* sp. from *Cannabis* in Lithuania and Iowa, respectively.

Most pathologists cite the causal agent of brown leaf spot as “*Phyllosticta cannabis*.” Among the 20 collections of “*Ph. cannabis*” examined from herb B and BPI, none proved to be a true *Phyllosticta* species (i.e., no conidial appendages). Most of these collections were accommodated in the genus *Phoma* (rarely, *Ascochyta*), based on in vivo morphology. Misidentified specimens of *Coniothyrium*, *Leptodothiorella*, *Fusicoccum* and *Phomopsis* spp. were also located (McPartland, 1992).

The most frequent cause of *Cannabis* brown leaf spot and stem canker is *Didymella cannabis* (anamorph *Phoma cannabis*), while *D. arcuata* (anamorph *A. arcuata*) is the second most frequently encountered. They are easily distinguished in vitro by the size, shape and septation of their conidia. Their teleomorphs also differ; *D. arcuata* ascospores are longer than those of *D. cannabis* by an average of 5 µm (37% larger). On the host, however, *P. cannabis*, *A. arcuata* and other species described in this paper are difficult to tell apart (TABLE 1).

At least three of the eight species, *P. exigua*, *P. piskorzii*, and *P. herbarum*, also attack a *Cannabis* relative, *Urtica dioica*. Morphologically similar nettles pathogens include *P. nebulosa* (Pers.: Fr.) Berk., *Ph. boehmeriicola* Davis, *P. hoehnellii* Kesteren, *A. urticae* Smith & Ramsbottom and *A. urticicola* Vanev & Bakalova. Two *Didymella* species on nettles parallel the relationship between *D. arcuata* and *D. cannabis*: *D. eupyrena* Sacc. has larger ascospores and conidia than *Didymella urticicola* van der Aa & Boerema (Boerema, 1976). These taxa await cross-inoculation studies with *D. cannabis* and *D. arcuata*.

ACKNOWLEDGMENTS

Curators of the following herbaria are thanked for loaning collections over the last 10 yr: B, BPI, BR, FH, MICH, ILLS and PAV. R. P. Korf provided ICBN interpretation. D. P. Rogers corrected the Latin. This work was partially supported by a grant from Hortapharm, B.V., The Netherlands.

LITERATURE CITED

- Boerema, G. H. 1970. Additional notes on *Phoma herbarum*. *Persoonia* 6: 15–48.
- . 1972. *Ascochyta phaseolorum* synonymous with *Phoma exigua*. *Netherlands J. Pl. Pathol.* 78: 113–115.
- . 1976. The *Phoma* species studied in culture by Dr. R. W. G. Dennis. *Trans. Brit. Mycol. Soc.* 67: 289–319.
- , and G. J. Bollen. 1975. Conidiogenesis and conidial septation as differentiating criteria between *Phoma* and *Ascochyta*. *Persoonia* 8: 111–144.

- , H. A. van Kesteren, and W. M. Loerakker. 1981. Notes on *Phoma*. *Trans. Brit. Mycol. Soc.* 77: 61–74.
- , and W. M. Loerakker. 1981. *Phoma piskorzii* (Petrak) comb. nov., the anamorph of *Leptosphaeria acuta* (Fuckel) P. Karst. *Persoonia* 11: 311–315.
- Brandenburger, W. 1985. *Parasitische Pilze an Gefäßpflanzen in Europa*. Gustav Fischer Verlag, Stuttgart, Germany. 1248 pp.
- Brunaud, P. 1899. Sphaeropsidees recoltees jusqu'a ce jour dans la Charente-Interieure. *Acad. Rochelle. Soc. Sci. Nat. Charente-Interieure* 26: 51–140.
- Brundza, K. 1933. Kai kurie parazitiniai grybeliai, surinkti Liefuvoje 1927–1932. *Zemės ūkio Akad. Leidinys* 1932: 199–208.
- Buchanan, P. K. 1987. A reappraisal of *Ascochyta* and *Ascochyta* (Coelomycetes). *Mycol. Pap.* 156: 1–83.
- Crane, J. L., and C. A. Shearer. 1991. A nomenclator of *Leptosphaeria* Cesati & de Notaris. *Illinois Nat. Hist. Surv. Bull.* 34: 195–355.
- Domsch, K. H., W. Gams, and T. H. Anderson. 1980. *Compendium of soil fungi*. Vol. 1. Academic Press, New York. 859 pp.
- Farr, D. F. 1991. *Septoria* species on *Cornus*. *Mycologia* 83: 611–623.
- Farr, M. L. 1973. *An annotated list of Spegazzini's fungus taxa*. J. Cramer, Berlin, Germany. 1661 pp.
- Fuller, W. J., and A. G. Norman. 1944. The nature of the flora on field-retting of hemp. *Soil Sci. Soc. Amer. Proc.* 9: 101–105.
- , and ———. 1945. Biochemical changes involved in the decomposition of hemp bark by pure cultures of fungi. *J. Bacteriol.* 50: 667–671.
- Greuter, A., et al. 1988. *International code of botanical nomenclature, 14th International Botanical Congress, Berlin*. Koeltz Scientific Books, Königstein, Germany. 328 pp.
- Kyokai, N.S.B. 1985. *Norin byogauchū meikan*. Japan Plant Protection Association, Tokyo. 412 pp.
- Lind, J. 1913. *Danish fungi*. Nordisk Forlag, Copenhagen, Denmark. 648 pp.
- McPartland, J. M. 1991. Common names for diseases of *Cannabis sativa* L. *Pl. Dis.* 75: 226–227.
- . 1992. The *Cannabis* pathogen project: report of the second five-year plan. *Mycol. Soc. Amer. Newslett.* 43(1): 43. (Abstract)
- . 1994a. *Cannabis* pathogens VIII: misidentifications appearing in the literature. *Mycotaxon* 53: 407–416.
- . 1994b. *Cannabis* pathogens IX: anamorphs of *Botryosphaeria* spp. *Mycotaxon* 53: 417–424.
- . 1994c. *Cannabis* pathogens XI: *Septoria* spp. on *Cannabis sativa*, *sensu stricto*. *Sydowia* 46: (In press)
- Morgan-Jones, G. 1988. Studies in the genus *Phoma*. XIV. Concerning *Phoma herbarum*, the type species, a widespread saprophyte. *Mycotaxon* 33: 81–90.
- , and K. B. Burch. 1988. Studies in the genus *Phoma*. XIII. Concerning *Phoma exigua* var. *exigua*, a cosmopolitan, ubiquitous fungus on diseased and dead plant material. *Mycotaxon* 32: 477–490.
- , M. R. Owsley, and J. M. Krupinsky. 1991. Notes on coelomycetes. IV. *Phyllosticta andropogonivora*, a causal organism of leafspot disease of *Schizachyrium scoparium* (little bluestem) and *Andropogon gerardii* (big and sand bluestem), reclassified in *Ascochyta*. *Mycotaxon* 42: 53–61.
- Müller, E., and J. A. von Arx. 1962. Die Gattungen der didymosporen Pyrenomyceten. *Beitr. Kryptogamenfl. Schweiz* 11(2): 1–922.
- Oudemans, C. A. J. A. 1920. *Enumeratio Systematica Fungorum* II. Comitum, The Hague, Netherlands.
- Paulsen, A. Q. 1971. Plant diseases affecting marijuana (*Cannabis sativa*). Unpubl. manuscript, Kansas State University, Manhattan. 11 pp.
- Petrak, F. 1923. Mykologische Notizen. *Ann. Mycol.* 21: 182–355.
- Röder, K. 1937. *Phyllosticta cannabis* (Kirchner?) Speg. eine Nebeufruchtform von *Mycosphaerella cannabis* (Winter) n.c. *Z. Pflanzenkrankh.* 47: 526–531.
- . 1939. Über einen neuen Hanfschädiger, *Didymella arcuata* n. sp. und seine Nebenfruchtformen. *Phytopathol. Z.* 12: 321–333.
- Saccardo, P. A. 1884. *Sylloge Fungorum omnium hucusque cognitorum* 3. 860 pp.
- , and C. Roumeguère. 1883. Reliquiae Libertianae (series III). Pyrenomyceteae. *Rev. Mycol. (Toulouse)* 5: 233–239.
- Schultes, R. E., W. M. Klein, T. Plowman, and T. E. Lockwood. 1974. *Cannabis*: an example of taxonomic neglect. *Bot. Mus. Leaflet* 23: 337–367.
- Shoemaker, R. A. 1984. Canadian and some extralimital *Leptosphaeria* species. *Canad. J. Bot.* 62: 2688–2729.
- Spegazzini, C. 1881. Nova addenda ad mycolgium Venetam. *Atti Soc. Crittog. Ital. Milano* 3: 42–71.
- Srivastara, S. L., and S. C. Naithani. 1979. *Cannabis sativa* Linn., a new host for *Phoma* sp. *Curr. Sci.* 48: 1004–1005.
- Stevenson, J. A. 1967. Rabenhorst and fungi exsiccati. *Taxon* 16: 112–119.
- Sutton, B. C. 1980. *The Coelomycetes-Fungi imperfecti with pycnidia, acervuli, and stromata*. Commonwealth Agricultural Bureau Publications, London, United Kingdom. 696 pp.
- , and J. M. Waterston. 1966. *Ascochyta phaseolorum*. C.M.I. descriptions of pathogenic fungi and bacteria No. 81. Commonwealth Agricultural Bureau Publications, London, United Kingdom.
- Sydow, P. 1898. Index universalis et locupletissimus nominum plantarum hospitem specierumque fungorum. *Sylloge Fungorum* 13.
- Voglino, P. 1913. I funghi piu dannosi alle piante osservati nella provincia di Torino e regioni vicine nel 1911. *Ann. Reale Accad. Agric. Torino* 55: 199–227.